

Chang Liu

TIME-DOMAIN ASTRONOMER | HINTZE FELLOW

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Education

PhD Northwestern University Evanston, US Jun 2026
BSc (Hons) Peking University Beijing, China Jun 2020

Research Interests

- **Survey Science & Machine Learning:** Leveraging advanced statistical and machine learning techniques to identify population-level patterns of transients in the Rubin/LSST era.
- **Supernovae (SNe):** Infant SNe, nebular-phase spectroscopy, spectral modeling.
- **Tidal disruption events (TDEs):** Repeating partial TDEs, exotic nuclear transients.
- **Software:** Open source software development, data reduction pipelines.

Skills

Astronomy Transient follow-up, spectroscopic observations and data reduction, hydrodynamical simulations
Data Science Hierarchical Bayesian modeling, Gaussian processes, gradient boosting
Programming **Proficient:** Python **Experienced:** Shell, Fortran, SQL

- HostSub_GP (**developer**): removing host galaxy background in transient long-slit spectroscopy using Gaussian processes and JAX acceleration

Softwares

- BayeSpecFit (**developer**): fitting blended SN spectral features with Bayesian inference
- Pypelt: optical/NIR spectrum reduction

Telescope Experience

Obs Runs Keck I (LRIS || 6.5 n, MOSFIRE || 0.5 n), Keck II (DEIMOS || 1.5 n, KCWI || 0.5 n) 2023 – 2026
Obs Runs Magellan Baade (FIRE || 3 n) 2025 – 2026
PI 2.56 m Nordic Optical Telescope (NOT), NOIRLab || 29 hr 2025A – 2026B
PI 4.1 m Southern Astrophysical Research (SOAR) Telescope, NOIRLab || 17 hr 2025A – 2026B
PI 6.5 m Magellan Baade Telescope, Northwestern || 3 n 2025A – 2026A
PI 10 m Keck Telescopes, Northwestern || 3.5 n 2024B – 2025B
co-I James Webb Space Telescope, NIRSpec + NIRCам (PI Goobar) || 10.2 hr DD 12764, Cycle 4
co-I James Webb Space Telescope, NIRSpec + MIRI (PI Jacobson–Galán) || 6 hr DD 9492, Cycle 4
co-I James Webb Space Telescope, NIRSpec + MIRI (PI Kwok) || 12.9 hr DD 6811, Cycle 3

Invited Talks & Seminars

Invited Talk Lorentz Center Workshop on Thermonuclear Explosions Leiden, Netherlands Jan 2027
Invited Talk ICESUN Workshop 2026: Type Ia Supernovae Wenshan, China Aug 2026
Seminar Carnegie arXiv Tea, Carnegie Observatories Pasadena, US Sep 2025
Seminar Astronomy Tea Talk, California Institute of Technology Pasadena, US Sep 2025
Seminar DESI Special Seminar Berkeley, US Jun 2025
Seminar University of California, Santa Cruz Santa Cruz, US Jun 2025
Invited Talk LS4 Team Meeting Evanston, US Mar 2025
Seminar Peking University Beijing, China Dec 2023
Seminar Shanghai Jiao Tong University Shanghai, China Jun 2023

Contributed Talks & Posters

Talk MMT Science Symposium	Tucson, US Apr 2026
Talk 247th AAS Meeting	Phoenix, US Jan 2026
Talk Keck Science Meeting	Los Angeles, US Sep 2025
Poster Open SkAI 2025	Chicago, US Sep 2025
Poster CIERA Fellows at 15	Evanston, US Aug 2025
Talk One Hundred Years of Supernova Science	Saltsjöbaden, Sweden Aug 2025
Talk Cosmic Lighthouses: Astrophysical and Cosmological Challenges with Type Ia Supernovae	Cambridge, UK Jul 2025
Poster Center for Decoding the Universe Annual Conference	Stanford, US Jun 2025
Poster CoDEX: Symposium for Computation and Data Intensive Research	Evanston, US Apr 2025
Poster Transients from Space	Baltimore, US Mar 2025
Poster Rise_Time 2024	West Lafayette, US Aug 2024
Talk 243rd AAS Meeting	New Orleans, US Jan 2024
Talk The 32nd Texas Symposium on Relativistic Astrophysics	Shanghai, China Dec 2023
Talk ZTF Science Meeting	Pasadena, US Oct 2023
Poster Keck Science Meeting	Berkeley, US Sep 2023
Talk SPOKEN-WERRD 2022 Symposium	Virtual Nov 2022
Talk ZTF Science Meeting	Evanston, US Oct 2022
Talk PKU-DoA Undergraduate Astronomy Symposium	Beijing, China Sep 2020

Honors & Awards

Oct 2026	Hintze Prize Fellowship , Hintze Centre for Astrophysical Surveys, the University of Oxford
Oct 2020	First prize & Lin-bridge Prize for Excellent Undergraduate Research , 2020 PKU-DoA Undergraduate Astronomy Symposium (endowed by Prof. Douglas Lin)
May 2020	Outstanding Graduate , College Graduate Excellence Award of Beijing
May 2020	Outstanding Graduate , College Graduate Excellence Award of Peking University
Jun 2019	Caltech Summer Undergraduate Research Fellowship (SURF) , California Institute of Technology
May 2019	National Innovation Training Program , Undergraduate Research & Training Program
Oct 2018	Wei Lin Scholarship , Annual scholarship of 2017-2018, Peking University
Oct 2017	Merit Student Pacesetter (the Utmost Annual Accolade at PKU) , Annual honor of 2016-2017, Peking University
Oct 2017	Arawana Scholarship , Annual scholarship of 2016-2017, Peking University

Professional Service

Referee	Astrophysical Journal
Referee	Astrophysical Journal Letters
Referee	Science Bulletin

Outreach

Lecturer	Evanston, US
RESEARCH EXPERIENCE IN ASTRONOMY AT CIERA FOR HIGH SCHOOL STUDENTS (REACH)	Jul 2023
Invited Speaker (<i>supernovae: from the past to the future</i>)	Beijing, China
PEKING UNIVERSITY YOUTH ASTRONOMY SOCIETY (PKU-YAS)	Apr 2021
Volunteer	Beijing, China
SUMMER CAMP OF ASTRONOMY FOR HIGH SCHOOL STUDENTS, PEKING UNIVERSITY	Jul 2018

References

Prof. Adam A. Miller

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Evanston, US

Department of Physics and Astronomy, Northwestern University

Prof. Enrico Ramirez-Ruiz

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Santa Cruz, US

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Prof. Xian Chen

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Beijing, China

Department of Astronomy, Peking University

Publication List

10 first-author papers (including 1 co-first authorship) among 45 total publications (h-index: 15)

First-author Publications

- **C. Liu** & A. A. Miller., Decoding the Early-Time Light Curves of Type Ia Supernovae. I. A Hierarchical Bayesian Framework for Demographic Inference, *arXiv: 2607.00075, submitted to ApJ*.
- **C. Liu**, A. A. Miller, N. Sarin, et al., Decoding the Early-Time Light Curves of Type Ia Supernovae. II. Population Parameters of One Thousand ZTF Supernovae, *arXiv: 2607.00081, submitted to ApJ*.
- **J. Hinkle**, **C. Liu**, A. A. Miller, et al., A Disappearing Act: Constraints From “Missing” Flares of Repeating Partial TDE Candidates, *arXiv: 2606.06578, submitted to ApJL*.
- **C. Liu** & A. A. Miller., HostSub_GP: Precise Galaxy Background Subtraction in Transient Spectroscopy with Gaussian Processes, 2026, *PASP*, 138, 024503.
- **C. Liu**, A. A. Miller, J. S. Bloom, et al., A Morphological Model to Separate Resolved–Unresolved Sources in the DESI Legacy Surveys: Application in the LS4 Alert Stream, 2025, *PASP*, 137, 084501.
- **C. Liu**, R. Yarza, & E. Ramirez-Ruiz., Repeating Partial Tidal Encounters of Sun-like Stars Leading to their Complete Disruption, 2025, *ApJ*, 979, 40.
- **C. Liu**, A. A. Miller, S. J. Boos, et al., SN 2022joj: A Peculiar Type Ia Supernova Possibly Driven by an Asymmetric Helium-shell Double Detonation, 2023, *ApJ*, 958, 178.
- **C. Liu**, A. A. Miller, A. Polin, et al., SN 2020jgb: A Peculiar Type Ia Supernova Triggered by a Helium-Shell Detonation in a Star-Forming Galaxy, 2023, *ApJ*, 946, 83.
- **C. Liu**, B. Mockler, E. Ramirez-Ruiz, et al., Tidal Disruption Events from Eccentric Orbits and Lessons Learned from the Noteworthy ASASSN-14ko, 2023, *ApJ*, 944, 184.
- **C. Liu**, X. Chen, & F. Du, Impact of an Active Sgr A* on the Synthesis of Water and Organic Molecules Throughout the Milky Way, 2020, *ApJ*, 899, 2.

Co-author Publications

- B. Barna et al., Tomography analysis of the intermediate-luminosity Type Iax SN 2024pxl, 2026, *A&A in press*.
- P. J. Pessi et al., SN 2024afyu interpreted as a Pair Instability Supernova, *submitted to A&A*.
- J. Johansson et al., Follow-up of SN 2025wny III: Spectroscopic Time-delay Measurements of a Strongly Gravitationally Lensed Superluminous Supernova, *submitted to ApJ*.
- M. L. Li et al., Follow-up of SN 2025wny II: Superluminous Supernova Physics at Cosmic Noon, *submitted to ApJ*.
- L. A. Kwok et al., JWST Spectroscopy of Type Ia Supernova 2025rbs from Maximum Light to the Nebular Phase, *submitted to ApJ*.
- C.-G. Touchard-Paxton et al., Peak ‘Nebular’ Emission and Early Flux Excesses in Ca-strong Supernovae, 2026, *A&A in press*.
- S. Belkin et al., GOTO identification and broadband modelling of the counterpart to the SVOM GRB 250818B, 2026, *MNRAS*, 550, 2.
- C. L. Ransome et al., The Eye of Sauron in SN 2025ngs: a Short-plateau Cousin of SN 1998S with Evidence for a Ring-like Circumstellar Medium, 2026, *ApJ*, 1006, 230.
- J. H. Terwel et al., An analysis of the Type Ia SN 2024gy and a comparison of different host extinction estimation techniques, 2026, *A&A*, 712, A173.

- T. E. Müller-Bravo et al., ZTF SN Ia DR2 follow-up: early excess in Type Ia supernova light curves, 2026, *A&A in press*.
- S. Hall et al., JWST Observations of Calcium-Strong Transients: I. Complex Nebular He Emission in SN 2024uj, *arXiv: 2607.00111*, submitted to *ApJ*.
- D. A. Perley et al., AT2024wpp: An Extremely Luminous Fast Ultraviolet Transient Powered by Accretion onto a Black Hole, 2026, *MNRAS*, 549, 1.
- P. Chen et al., Detection of persistent helium absorption in the 91bg-like type Ia Supernova 2022an, 2026, *ApJL*, 1004, L42.
- R. Hwangbo et al., Near-Infrared and Optical Observations of SN 2024rbc: The First Early Detection of CO and Dust in a Type Ib Supernova, 2026, *ApJ in press*.
- A. Singh et al., Photometry and Spectroscopy of SN 2024pxl: A Luminosity Link Among Type Iax Supernovae, 2026, *ApJ*, 999, 227.
- C. Sevilla et al., Multiwavelength Analysis of Six Luminous, Fast Blue Optical Transients, 2026, *ApJ in press*.
- M. M. Kasliwal et al., ZTF25abjmnps (AT2025ulz) and S250818k: A Candidate Superkilonova from a Sub-threshold Sub-Solar Gravitational Wave Trigger, 2025, *ApJL*, 995, L59.
- J. Pearson et al., Mid-Infrared Dust Evolution and Late-time Circumstellar Medium Interaction in SN 2017eaw, 2025, *ApJ*, 993, 213.
- A. C. Gordon et al., Mapping the Spatial Distribution of Fast Radio Bursts within their Host Galaxies, 2025, *ApJ*, 993, 119.
- A. A. Miller et al., The La Silla Schmidt Southern Survey, 2025, *PASP*, 137, 094204.
- L. A. Kwok, **C. Liu**, S. W. Jha, et al., JWST Spectroscopy of SN Ia 2022aaq and 2024gy: Evidence for Enhanced Central Stable Ni Abundance and a Deflagration-to-Detonation Transition, 2026, *ApJ*, 1005, 52.
- P. J. Pessi et al., The ambiguous AT2022rze: Changing-look AGN mimicking a supernova in a merging galaxy system, 2025, *MNRAS*, 542, 4.
- L. A. Kwok et al., JWST and Ground-based Observations of the Type Iax Supernovae SN 2024pxl and SN 2024vjm: Evidence for Weak Deflagration Explosions, 2025, *ApJL*, 989, L33.
- A. Y. Q. Ho et al., A Luminous Red Optical Flare and Hard X-ray Emission in the Tidal Disruption Event AT2024kmq, 2025, *ApJ*, 989, 54.
- J. C. Rastinejad et al., EP 250108a/SN 2025kg: Observations of the most nearby Broad-Line Type Ic Supernova following an Einstein Probe Fast X-ray Transient, 2025, *ApJL*, 988, L13.
- Y. Yao et al., A Massive Black Hole 0.8 kpc from the Host Nucleus Revealed by the Offset Tidal Disruption Event AT2024tvd, 2025, *ApJL*, 985, L48.
- N. Rehemtulla et al., The BTSbot-nearby discovery of SN 2024jlf: rapid, autonomous follow-up probes interaction in an 18.5 Mpc Type IIP supernova, 2025, *ApJ*, 985, 241.
- K. Das et al., Low-Luminosity Type IIP Supernovae from the Zwicky Transient Facility Census of the Local Universe. I: Luminosity Function, Volumetric Rate, 2025, *PASP*, 137, 044203.
- L. Harvey et al., ZTF SN Ia DR2: High-velocity components in the Si II λ 6355, 2025, *A&A*, 695, A264.
- T. Eftekhari et al., The Massive and Quiescent Elliptical Host Galaxy of the Repeating Fast Radio Burst FRB 20240209A, 2025, *ApJL*, 979, L22.
- G. Dimitriadis et al., ZTF SN Ia DR2: The diversity and relative rates of the thermonuclear supernova population, 2024, *A&A*, 694, A10.
- Z. Wu et al., Gaia22dkvLb: A Microlensing Planet Potentially Accessible to Radial-Velocity Characterization, 2024, *AJ*, 168, 62.
- K. Das et al., SN 2023zaw: an ultra-stripped, nickel-poor supernova from a low-mass progenitor, 2024, *ApJL*, 969, L11.
- P. Chen et al., A 12.4 Day Periodicity in a Close Binary System after a Supernova, 2023, *Nature*, 625, 7994, 253-258.
- G. Dimitriadis et al., SN 2021zny: an early flux excess combined with late-time oxygen emission suggests a double white dwarf merger event, 2023, *MNRAS*, 521, 1162.